

Nxt Assess - Project Overview

1. Project Features

Core Functionalities:

- Secure Authentication: JWT-based login with credential validation
- Timed Assessments: 10-minute timer with auto-submission
- Multiple Question Types:
 - Text-based options (DEFAULT)
 - Image-based options (IMAGE)
 - Single-select dropdown (SINGLE_SELECT)
- Real-time Tracking:
 - Answered/Unanswered counters
 - Question navigation with visual indicators
- Results Dashboard: Score breakdown + time taken analytics

UI/UX Highlights:

- Responsive design (mobile/tablet/desktop)
- Interactive question navigation
- Visual feedback for selected options

2. Challenges & Exciting Experiences

Technical Challenges:

- Dynamic Question Rendering:
 - Challenge: Handling 3 distinct question types in a single flow
 - Solution: Created a Question component with switch-case rendering
- Timer-State Synchronization:
 - Challenge: Preventing memory leaks with countdown
 - Solution: Used useEffect cleanup + global state
- Authentication Flow:
 - Challenge: Redirecting authenticated users from login
 - Solution: Implemented ProtectedRoute wrapper

Exciting Moments:

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- Successfully implementing auto-submit on timeout
- Building the image-based question interface
- Achieving 100% test coverage for critical flows

3. Tools & Technologies

Frontend:

- React.js (Components, Hooks, Context)
- React Router (Navigation)
- Fetch API (HTTP requests)
- CSS Modules (Styling)

Dev Tools:

- Figma (UI Design)
- Postman (API Testing)
- Git (Version Control)
- ESLint (Code Quality)

Testing:

- Jest + React Testing Library
- Data-testid attributes for UI testing

4. Industry Best Practices Applied

Security:

- Password masking + secure JWT storage
- Protected routes for authenticated content

Performance:

- Lazy loading for routes
- Memoization of question components

Accessibility:

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- Semantic HTML (button vs div)
- ARIA labels for interactive elements
- Keyboard navigation support

Code Quality:

- Modular component architecture
- PropTypes for type checking
- Error boundaries for graceful failures

5. Development Approach

Methodology: Agile Sprints

Key Phases:

1. Planning:

- Figma prototyping
- API contract finalization

2. Implementation:

- Auth flow -> Assessment engine -> Results
- Mobile-first responsive design

3. Testing:

- Unit tests for business logic
- Integration tests for user flows

Collaboration:

- Daily standups (for personal accountability)
- Git feature branches

6. Future Enhancements & Implementation Plan

Planned Features:

- Multi-select Questions: New MULTI_SELECT option_type + checkbox UI (High)
- Leaderboard: Backend API + React-Table integration (Medium)

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- Offline Mode: Service Workers + IndexedDB caching (Low)
- PDF Reports: React-PDF library + score analytics (Medium)

Technical Debt Resolution:

1. State Management:

Current: Component state -> Future: Redux Toolkit

2. API Mocking:

Current: Static JSON -> Future: MirageJS simulation

3. Testing:

Current: Unit tests -> Future: Cypress E2E tests

Note: Gantt chart and visuals can be presented separately if required.

New Feature:

- Review Failed Questions: Allow users to revisit incorrectly answered questions post-assessment with explanations (High)

Implementation: Add a review screen that pulls incorrect answers and displays correct responses + optional explanation tooltip.

7. Application Working Flow

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1. User Authentication:

- User enters credentials on login screen
- Credentials are validated using backend API
- On success, token is stored in cookies and user is redirected to assessment

2. Assessment Initialization:

- Assessment page loads with question set (from static or API)

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- 10-minute timer starts using useEffect hook

3. Question Navigation & Answering:

- Users navigate between questions using a sidebar or next/prev buttons
- Answer selection is stored in local component state or context

4. Timer & Auto Submission:

- When timer ends, auto-submit logic is triggered using useEffect cleanup
- Answers are collected and scored based on correct responses

5. Results Display:

- User sees their score, total questions, correct/incorrect breakdown
- Option to review failed questions (future enhancement)

6. Logout:

- User can log out to clear the cookie and return to login screen

Sample User Credentials

- aakash / sky@007
- agastya / myth#789
- advika / world@5
- binita / modest*6
- chetan / vigor\$life
- deepak / lightstar@1
- harshad / joy@85
- kapil / moon\$008
- rahul / rahul@2021
- shravya / musical#stone
- saira / princess@9